LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 4

Course Code: CSE276
Course Title: ARTIFICIAL INTELLIGENCE FOUNDATIONS
Course Outcome: CO4
Paper: Sample Subjective Questions
Maximum Marks: 10 x 10 = 100

Unit: 4
Unit Title: Evolution of Modern AI and Foundation Models
Unit Mapped RBT: L3 (Apply)
Total Questions: 10
Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Apply the three types of machine learning to three tasks drawn from a retail business. (10 Marks)

(a) State one task suited to each of supervised, unsupervised and reinforcement learning.

(b) Use the training signal each type receives to justify why each task belongs where you have placed it.

[L3 - Apply]

Q2. Compare symbolic Artificial Intelligence with statistical Artificial Intelligence. (10 Marks)

(a) State how each of the two obtains the rules by which it behaves.

(b) Use a named task to show what the statistical approach makes possible that the symbolic approach cannot.

[L3 - Apply]

Q3. Apply Edge Artificial Intelligence to a camera that must detect a safety breach on a factory floor. (10 Marks)

(a) State what runs on the device and what, if anything, is sent to a server.

(b) Use latency, privacy and connectivity to justify the design, and state what has to be given up.

[L3 - Apply]

Q4. Examine why deep learning became practical only in the last two decades although the ideas are far older. (10 Marks)

(a) State the two developments that made large networks trainable.

(b) Use this to explain why deep learning removed much of the need for hand-designed features.

[L3 - Apply]

Q5. Use the distinction between a foundation model and a task-specific model to plan a small project. (10 Marks)

(a) State what each of the two is trained on and what each costs to produce.

(b) Apply the distinction to a request for an assistant that answers questions about one department's procedures.

[L3 - Apply]

Q6. Apply quantisation and pruning in order to fit a model onto a mobile device. (10 Marks)

(a) State what each of the two does to the network.

(b) Use them together to describe how you would reduce a model, and state what you would measure afterwards.

[L3 - Apply]

Q7. Compare a convolutional network with a recurrent network. (10 Marks)

(a) State the structure of the data each of the two is designed for, and how each exploits that structure.

(b) Apply the comparison to two named tasks, assigning one architecture to each.

[L3 - Apply]

Q8. Examine an agentic Artificial Intelligence system and describe how it differs from an assistant that answers once and stops. (10 Marks)

(a) State what an agentic system does with a request before it produces a final answer.

(b) Use an example task to show where the extra capability helps, and state one risk it introduces.

[L3 - Apply]

Q9. Apply multimodal Artificial Intelligence to a task that combines images with text. (10 Marks)

(a) State the task and describe what the model receives as input.

(b) Compare the multimodal approach with a pipeline of separate single-modality models for the same task.
[L3 - Apply]

Q10. Demonstrate how Tiny Artificial Intelligence differs from an ordinary deep learning deployment. (10 Marks)

(a) State the hardware Tiny Artificial Intelligence targets and the limits that hardware imposes.
(b) Use a named application to show what has to be changed about a model before it will run there.
[L3 - Apply]

----- End of Sample Question Set -----